Lecture 7: Parameter Estimation Using Data Assimilation

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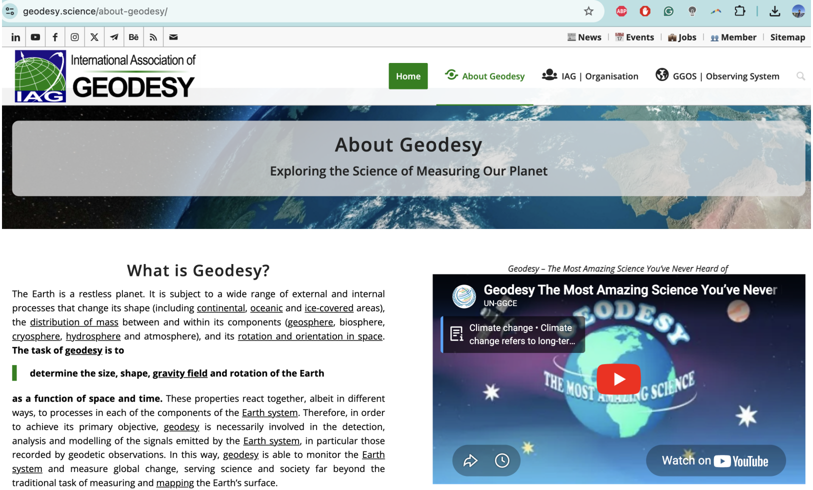
Introduction to Data Assimilation
2026 Spring

清華大學
Tsinghua University

Geodesy and DA Summer School Opportunity



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About Geodesy

Exploring the Science of Measuring Our Planet

What is Geodesy?

The Earth is a restless planet. It is subject to a wide range of external and internal processes that change its shape (including continental, oceanic and ice-covered areas), the distribution of mass between and within its components (geosphere, biosphere, cryosphere, hydrosphere and atmosphere), and its rotation and orientation in space. **The task of geodesy is to**

determine the size, shape, gravity field and rotation of the Earth

as a function of space and time. These properties react together, albeit in different ways, to processes in each of the components of the Earth system. Therefore, in order to achieve its primary objective, geodesy is necessarily involved in the detection, analysis and modelling of the signals emitted by the Earth system, in particular those recorded by geodetic observations. In this way, geodesy is able to monitor the Earth system and measure global change, serving science and society far beyond the traditional task of measuring and mapping the Earth's surface.

Geodesy – The Most Amazing Science You've Never Heard of

Geodesy and DA Summer School Opportunity

Discover Geodesy – The Science Behind Measuring and Monitoring Our World

What Geodesy Measures and Why It Matters?

Geodesy involves measuring and understanding the Earth's geometric and physical properties, including

- **Determining exact locations** (latitude, longitude and altitude)
- **Developing coordinate systems** (reference frames) for mapping, navigation and Earth observation
- Monitoring the **Earth's gravity field**
- Measuring **variations of the Earth rotation**
- Tracking the **movement of tectonic plates and surface deformation**
- Measuring **sea level changes**, **ice melting** and global water cycle
- Detecting environmental changes

Geodesy is essential for satellite positioning applications such as GNSS (Global Navigation Satellite Systems), observation of the Earth from satellites, land management and climate studies. It uses space and ground-based technologies to ensure accurate measurements.

Geodesy and DA Summer School Opportunity

Summer School: Satellite-Based Hydrological Data Assimilation

COM2, ECS, GGOS, ICCC, ICGEM, 📅 EVENT UPDATE

Registration is open for the Summer School of "SATELLITE-BASED HYDROLOGICAL DATA ASSIMILATION", August 25-28, 2026, which will be hosted by the Geodesy Research Group of Aalborg University, Denmark.

> Please register here; and the deadline is end of April 2026

The school is free of charge, but a registration process will be followed to select the most relevant audiences.

This Summer School offers early career researchers interested in large-scale hydrology research an opportunity to engage in dedicated lectures and practical sessions. The focus will be on advanced techniques for measuring and modeling the water cycle, as well as integrating Satellite Gravity, Satellite Altimetry, and Soil Remote Sensing into large-scale hydrological models. Participants will be introduced to ensemble-based sequential Data Assimilation (DA) techniques for merging satellite data with models. The school emphasizes hands-on learning, featuring practical exercises in Python, along with various interactive activities. The school emphasizes hands-on learning, featuring practical exercises in Python, along with various interactive activities.

Outline

1. Non-Bayesian methods
2. Bayesian methods (data assimilation, Markov chain Monte Carlo, etc)

Introduction: Different Parameter Estimation Methods

Roughly speaking, two categories:

1. Bayesian, and
2. Non-Bayesian

The underlying principle of Bayesian methods,

$$p(\boldsymbol{\theta}|\mathcal{D}) \propto p(\mathcal{D}|\boldsymbol{\theta})p(\boldsymbol{\theta})$$

which is used explicitly or implicitly in the parameter estimation.

- $\boldsymbol{\theta}$ is the collection of parameters
- $\mathcal{D}$ is the observational data
- posterior $\propto$ likelihood $\times$ prior

Non-Bayesian Methods

1. Regression

Example:

$$y = ax + bx^2 + cx^3 + f + \sigma\epsilon, \qquad \epsilon \sim \mathcal{N}(0,1).$$

Assume we have N data points (x_i, y_i) for $i = 1, \ldots, N$. The estimation of parameters $(a, b, c, f)^T$ can be carried out using standard regression method:

$$\begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_{N-1} \\ y_N \end{pmatrix} = \begin{pmatrix} x_1 & x_1^2 & x_1^3 & 1 \\ x_2 & x_2^2 & x_2^3 & 1 \\ \vdots & \vdots & \vdots & \vdots \\ x_{N-1} & x_{N-1}^2 & x_{N-1}^3 & 1 \\ x_N & x_N^2 & x_N^3 & 1 \end{pmatrix} \begin{pmatrix} a \\ b \\ c \\ f \end{pmatrix}$$

The parameters are given by the least square solution. The parameter σ can be computed via the residual variance.

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Question: Can this be easily applied to SDE?

Consider a simple example:

$$\mathrm{d}x = (-ax + f)\,\mathrm{d}t + \sigma\,\mathrm{d}W_x$$

the discrete version of which is

$$\frac{x_{i+1} - x_i}{\Delta t} = (-ax_i + f) + \frac{\sigma}{\sqrt{\Delta t}}\epsilon_i.$$

What is the issue here?

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What is the issue here?

$$\lim_{\Delta t \to 0} \frac{\sigma}{\sqrt{\Delta t}} \to \infty$$

2. Matching statistics

This example can be very useful for certain dynamics.

Example 1.

$$\mathrm{d}x = (-ax + f)\,\mathrm{d}t + \sigma\,\mathrm{d}W_x$$

Using a time series of $x_i, i = 1, \dots, N$, assuming N is sufficiently large, it is easy to compute the following three statistical quantities,

- μ: mean of x
- R: variance of x
- τ: decorrelation time of x

The three parameters a, f and σ are determined by these three quantities,

- $f/a = \mu$
- $\sigma^2/(2a) = R$
- $1/a = \tau$

Example 2.

$$dx = [f + ax + bx^2 + cx^3]dt + \sigma dW_A,$$

the associated PDF is

$$p_{eq}(x) = N_0 \exp\left(\frac{2}{\sigma^2}(fx + \frac{a}{2}x^2 + \frac{b}{3}x^3 + \frac{c}{4}x^4)\right).$$

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Can we make use of a long time series of x to form p_{eq} and then make use of the above formula to estimate the parameters?

Answer: Sometimes yes, sometimes no.

3. General: Choosing parameters to optimize a certain cost function.

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Optimization problem:

$$\boldsymbol{\theta} = \arg\min G(\boldsymbol{\theta})$$

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examples of the cost function $G(\boldsymbol{\theta})$:

1. a combination of the error in the equilibrium PDF and the temporal ACF
2. maximizing the prediction skill of f at a certain lead time
3. capturing certain features of nature
4. ...

4. Using physical knowledge.

This occurs usually in the dynamical model of physics.

The parameters are not actually “estimated”, but are inferred from physical understanding or observations.

Examples:

- In $F = ma$, if F is the gravity forcing, then $a = g = 9.8m/s^2$ is inferred from physical knowledge.

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Examples:

- In $F = ma$, if F is the gravity forcing, then $a = g = 9.8m/s^2$ is inferred from physical knowledge.

- The ocean current is forced by atmosphere wind

$$\frac{dU}{dt} = \text{shallow water equation} + \tau,$$

  where $\tau = \gamma u$. Here U is the ocean current while u is the atmosphere wind. Clearly the linear relationship $\tau = \gamma u$ is an approximation. The parameter γ is estimated based on the observational data based on simple linear regression technique. Such a simplified relationship is typically called **parameterization**.

Parameter Estimation Using Data Assimilation

Recall the conditional Gaussian systems,

$$\begin{aligned} d\mathbf{u}_{\mathbf{I}} &= [\mathbf{A}_0(t,\mathbf{u}_{\mathbf{I}}) + \mathbf{A}_1(t,\mathbf{u}_{\mathbf{I}})\mathbf{u}_{\mathbf{II}}]dt + \boldsymbol{\Sigma}_{\mathbf{I}}(t,\mathbf{u}_{\mathbf{I}})d\mathbf{W}_{\mathbf{I}}(t) \\ d\mathbf{u}_{\mathbf{II}} &= [\mathbf{a}_0(t,\mathbf{u}_{\mathbf{I}}) + \mathbf{a}_1(t,\mathbf{u}_{\mathbf{I}})\mathbf{u}_{\mathbf{II}}]dt + \boldsymbol{\Sigma}_{\mathbf{II}}(t,\mathbf{u}_{\mathbf{I}})d\mathbf{W}_{\mathbf{II}}(t) \end{aligned}$$

The conditional Gaussian system also provides a framework for parameter estimation. In fact, $\mathbf{u}_{\mathbf{II}}$ can be written as

$$\mathbf{u}_{\mathbf{II}} = (\widetilde{\mathbf{u}}_{\mathbf{II}}, \boldsymbol{\Lambda}),$$

where $\mathbf{u}_{\mathbf{II}}$ in $\mathbb{R}^{\widetilde{N}_2}$ is physical process variables and $\boldsymbol{\Lambda} = (\lambda_1, \lambda_2, \ldots, \lambda_p) \in \mathbb{R}^{N_{2,p}}$ denotes the model parameters. Here $N_2 = \widetilde{N}_2 + N_{2,p}$. Rewriting the conditional Gaussian system (1) in terms of $\mathbf{u}_{\mathbf{II}} = (\widetilde{\mathbf{u}}_{\mathbf{II}}, \boldsymbol{\Lambda})$ yields

$$\begin{aligned} d\mathbf{u}_{\mathbf{I}} &= [\mathbf{A}_0(t,\mathbf{u}_{\mathbf{I}}) + \mathbf{A}_1(t,\mathbf{u}_{\mathbf{I}})\widetilde{\mathbf{u}}_{\mathbf{II}} + \mathbf{A}_{1,\lambda}(t,\mathbf{u}_{\mathbf{I}})\boldsymbol{\Lambda}]dt + \boldsymbol{\Sigma}_{\mathbf{I}}(t,\mathbf{u}_{\mathbf{I}})d\mathbf{W}_{\mathbf{I}}(t), \\ d\mathbf{u}_{\mathbf{II}} &= [\mathbf{a}_0(t,\mathbf{u}_{\mathbf{I}}) + \mathbf{a}_1(t,\mathbf{u}_{\mathbf{I}})\widetilde{\mathbf{u}}_{\mathbf{II}} + \mathbf{a}_{1,\lambda}(t,\mathbf{u}_{\mathbf{I}})\boldsymbol{\Lambda}]dt + \boldsymbol{\Sigma}_{\mathbf{II}}(t,\mathbf{u}_{\mathbf{I}})d\mathbf{W}_{\mathbf{II}}(t). \end{aligned}$$

Consider the parameter estimation in the following simple setup:

$$d\mathbf{u}_{\mathbf{I}} = [\mathbf{A}_0(t, \mathbf{u}_{\mathbf{I}}) + \mathbf{A}_{1,\lambda}(t, \mathbf{u}_{\mathbf{I}})\boldsymbol{\Lambda}^*]dt + \boldsymbol{\Sigma}_{\mathbf{I}}(t, \mathbf{u}_{\mathbf{I}})d\mathbf{W}_{\mathbf{I}}(t).$$

Given the observed trajectory $\mathbf{u}_{\mathbf{I}}$, our goal is to estimate the parameter $\boldsymbol{\Lambda}^*$.

Note: There are different parameter estimation methods.

1. Direct parameter estimation algorithm.

Since $\mathbf{\Lambda}$ are constant parameters, it is natural to augment the dynamics with the following relationship,

$$d\mathbf{u_I} = [\mathbf{A}_0(t, \mathbf{u_I}) + \mathbf{A}_{1,\lambda}(t, \mathbf{u_I})\mathbf{\Lambda}]dt + \mathbf{\Sigma_I}(t, \mathbf{u_I})d\mathbf{W_I}(t), \tag{15a}$$

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d\mathbf{\Lambda} &= 0, && \text{(15b)}
\end{aligned}
$$

where an initial uncertainty for the parameter $\mathbf{\Lambda}$ is assigned.

The time evolutions of the mean $\overline{\mathbf{u}}_{\mathbf{II}}$ and covariance $\mathbf{R_{II}}$ of the estimate of $\mathbf{\Lambda}$ are given by

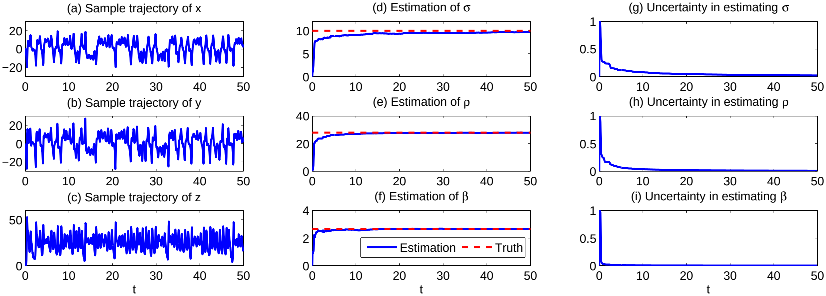
$$
\begin{aligned}
d\overline{\mathbf{u}}_{\mathbf{II}}(t) =& (\mathbf{R_{II}}\mathbf{A}_1^*(t,\mathbf{u_I}))(\mathbf{\Sigma_I}\mathbf{\Sigma_I}^*)^{-1}(t,\mathbf{u_I})\left[d\mathbf{u_I} - (\mathbf{A}_0(t,\mathbf{u_I}) + \mathbf{A}_1(t,\mathbf{u_I})\overline{\mathbf{u}}_{\mathbf{II}})dt\right], && \text{(16a)}\\
d\mathbf{R_{II}}(t) =& -(\mathbf{R_{II}}\mathbf{A}_1^*(t,\mathbf{u_I}))(\mathbf{\Sigma_I}\mathbf{\Sigma_I}^*)^{-1}(t,\mathbf{u_I})(\mathbf{R_{II}}\mathbf{A}_1^*(t,\mathbf{u_I}))^*dt. && \text{(16b)}
\end{aligned}
$$

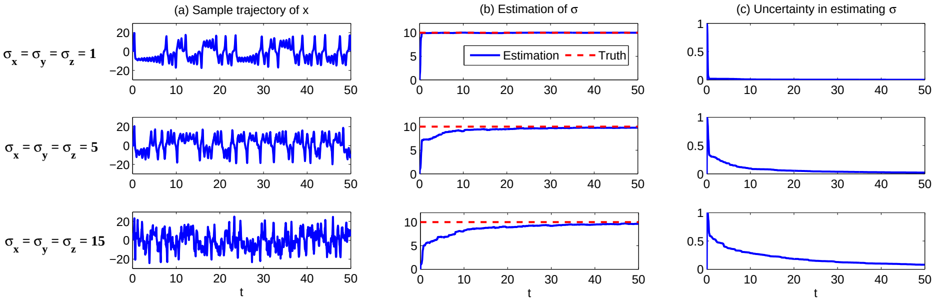
The formula in (16b) indicates that $\mathbf{R_{II}} = 0$ is a solution, plugging which into (16a) results in $\overline{\mathbf{u}}_{\mathbf{II}} = \mathbf{\Lambda}^*$. This means by knowing the perfect model the estimated parameters in (15)–(16) under certain conditions will converge to the truth.

As a simple test example, consider estimating the three parameters σ, ρ and γ in the noisy L-63 model with $\rho = 28, \sigma = 10, \beta = 8/3$.

$$
\begin{aligned}
dx &= \sigma(y - x)dt + \sigma_x dW_x, \\
dy &= \big(x(\rho - z) - y\big)dt + \sigma_y dW_y, \\
dz &= (xy - \beta z)dt + \sigma_z dW_z, \\
d\sigma &= 0, \\
d\rho &= 0, \\
d\beta &= 0,
\end{aligned}
$$

with $\sigma_x = \sigma_y = \sigma_z = 5$.




(a) Sample trajectory of x
(b) Estimation of σ
(c) Uncertainty in estimating σ
σx = σy = σz = 1
σx = σy = σz = 5
σx = σy = σz = 15
Estimation
Truth
t
t
t

2. Parameter estimation using stochastic parameterized equation.

A new approach of the augmented system can be formed in the following way:

$$d\mathbf{u_I} = [\mathbf{A}_0(t, \mathbf{u_I}) + \mathbf{A}_{1,\lambda}(t, \mathbf{u_I})\mathbf{\Lambda}]dt + \mathbf{\Sigma_I}(t, \mathbf{u_I})d\mathbf{W_I}(t), \tag{17a}$$

$$d\mathbf{\Lambda} = [\mathbf{c}_1\mathbf{\Lambda} + \mathbf{c}_2]dt + \boldsymbol{\sigma}_{\mathbf{\Lambda}} d\mathbf{W}_{\mathbf{\Lambda}}(t). \tag{17b}$$

Here, $\mathbf{c}_1$ is a negative-definite diagonal matrix, $\mathbf{c}_2$ is a constant vector and $\boldsymbol{\sigma}_{\mathbf{\Lambda}}$ is a diagonal noise matrix.

- The stochastic parameterized equations in (17b) serve as the prior information of the parameter estimation.

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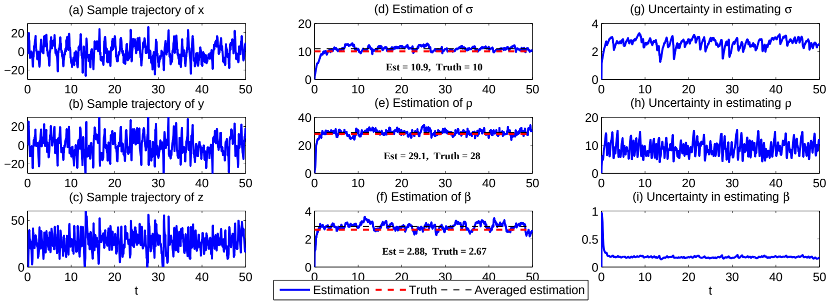
- The stochastic parameterized equations in (17b) serve as the prior information of the parameter estimation.
- Although certain model error will be introduced in the stochastic parameterized equations due to the appearance of $\mathbf{c}_1$, $\mathbf{c}_2$ and $\boldsymbol{\sigma}_{\mathbf{\Lambda}}$, it has shown that the convergence rate will be greatly accelerated.
- In fact, in linear models, rigorous analysis reveals that the convergence rate using stochastic parameterized equations (17) is exponential while that using the direct method (17) is only algebraic.

Now we apply the parameter estimation using stochastic parameterized equations (17) for the noisy L-63 model with a large noise $\sigma_x = \sigma_y = \sigma_z = 15$. The augmented system reads,

$$
\begin{aligned}
dx &= \sigma(y - x)dt + \sigma_x dW_x, \\
dy &= \big(x(\rho - z) - y\big)dt + \sigma_y dW_y, \\
dz &= (xy - \beta z)dt + \sigma_z dW_z, \\
d\sigma &= -d_\sigma(\sigma - \hat{\sigma})dt + \sigma_\sigma dW_\sigma, \\
d\rho &= -d_\rho(\rho - \hat{\rho})dt + \sigma_\rho dW_\rho, \\
d\beta &= -d_\beta(\beta - \hat{\beta})dt + \sigma_\beta dW_\beta.
\end{aligned}
$$



Bayesian Methods

1. data assimilation
2. maximum likelihood
3. maximum a posteriori
4. other cost function within the Bayesian framework

If the prior is non-informative, then 2) and 3) are equivalent.

Most of the Bayesian optimization methods are local method, since the cost function is usually not convex.

One global method that is widely used is called “MCMC” (Markov chain Monte Carlo).

Parameter estimation of linear Gaussian models is well understood. Analytic expressions are available for likelihood, posterior and many other cost functions.

For nonlinear/non-Gaussian models

- closed analytic formulae are often not available
- high dimensionality makes the convergence extremely slowly
- Partial observations lead to a huge difficulty in sampling hidden trajectories.

Markov chain Monte Carlo (MCMC)

Markov chain Monte Carlo (MCMC) methods comprise a class of algorithms for sampling from a probability distribution.

As in the name, it has two components:

- Markov chain
- Monte Carlo

Monte Carlo

Monte Carlo methods, or Monte Carlo experiments, are a broad class of computational algorithms that rely on repeated random sampling to obtain numerical results.

Example 1. Computing the integral

$$I = \int_0^1 x^2 dx$$

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for i = 1:5
    N = 10^i;
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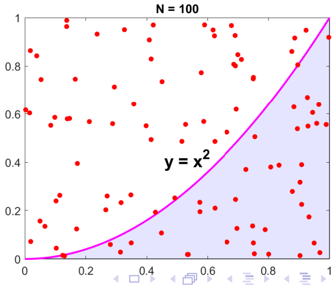
end
```

Answers are

$$\begin{aligned} N &= 10, & I &= 0.2000 \\ N &= 100, & I &= 0.4000 \\ N &= 1000, & I &= 0.3330 \\ N &= 10000, & I &= 0.3285 \\ N &= 100000, & I &= 0.3346 \end{aligned}$$



Example 2. Computing π.

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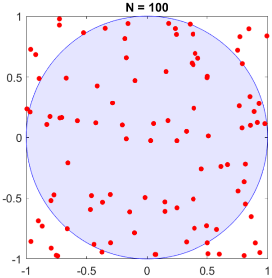
```
for i = 1:5
    N = 10^i;
    x = 2 * rand(1,N) - 1; y = 2 * rand(1,N) - 1;
    area = 2 * 2;
    I = sqrt(x.^2 + y.^2) - 1; I = sum(I < 0) / N * area;
    disp(I);
end
```

Answers are

$$N = 10, \qquad I = 2.2000$$
$$N = 100, \qquad I = 2.9200$$
$$N = 1000, \qquad I = 3.1120$$
$$N = 10000, \qquad I = 3.1520$$
$$N = 100000, \qquad I = 3.1386$$



Markov Chain

A Markov chain is a stochastic model describing a sequence of possible events in which the probability of each event depends only on the state attained in the previous event.

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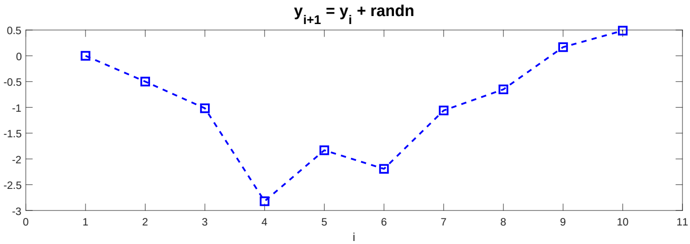
Example:

$$y_{i+1} = y_i + \epsilon, \qquad \epsilon \sim \mathcal{N}(0, 1)$$



Metropolis-Hastings algorithm

- Metropolis-Hastings algorithm is one of the most widely used MCMC algorithm.
- It is a method for obtaining a sequence of random samples from a probability distribution from which direct sampling is difficult.
- Metropolis algorithm requires to use a symmetric proposal distribution while no such a requirement is in the Metropolis-Hastings.

The Metropolis algorithm.

- Let $f(x)$ be a function that is proportional to the desired probability distribution $P(x)$ (a.k.a. a target distribution).
- Let g be a symmetric density function, namely $g(x|y) = g(y|x)$. A common choice of g is a Gaussian distribution centered at y. The function g is referred to as the proposal density.
- Initialization: Choose an arbitrary point x_t to be the first sample and choose a proposal density $g(x|y)$ that suggests a candidate for the next sample value x, given the previous sample value y so that points closer to y are more likely to be visited next, making the sequence of samples into a random walk.
- For each iteration t:
  1. Generate a candidate x' for the next sample by picking from the distribution $g(x'|x_t)$.
  2. Calculate the acceptance ratio $\alpha = f(x')/f(x_t)$, which will be used to decide whether to accept or reject the candidate. Because f is proportional to the density of P, we have that $\alpha = f(x')/f(x_t) = P(x')/P(x_t)$
  3. Accept or reject:
     - 3.1 Generate a uniform random number $u \in [0, 1]$
     - 3.2 If $u \leq \alpha$, then accept the candidate by setting $x_{t+1} = x'$.
     - 3.3 If $u > \alpha$, then reject the candidate and set $x_{t+1} = x_t$ instead.

In Metropolis-Hastings algorithm

$$\alpha = \frac{f(x')}{f(x_t)} = \frac{P(x')g(x|x')}{P(x_t)g(x'|x)}$$

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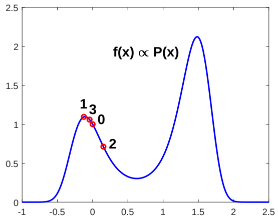
Example. Use the Metropolis algorithm to sample from the density function $p(x)$, where

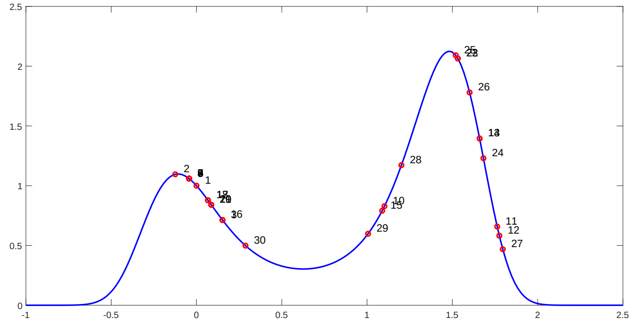
$$p(x) \propto \exp\left((-3.2x - 11.2x^2 + 21.33x^3 - 8x^4)\right)$$

Let $g(x'|x_t) = (2\pi)^{-1/2} \exp((x' - x_t)^2/2)$.
Start with $x_0 = 0$.

t	x'	α	u	x_t	
1	-0.1242	1.0952	0.0259	-0.1242	accept
2	0.1530	0.7119	0.4353	0.1530	accept
3	-0.0432	1.0595	0.3303	-0.0432	accept
4	-1.1722	0.0000	0.6193	-0.0432	reject
5	-0.6503	0.0069	0.2668	-0.0432	reject



First 30 sample points.



0.0000	-0.1242	0.1530	-0.0432	-0.0432	-0.0432	-0.0432
-0.0432	-0.0432	1.1020	1.7630	1.7754	1.6599	1.6599
1.0884	0.1520	0.0674	0.0674	0.0866	0.0866	0.0866
1.5321	1.5321	1.6822	1.5195	1.6012	1.7957	1.2013
1.0059	0.2873					

The acceptance rate is 45% in this case. The acceptance rate depends on the proposal.

Why need MCMC?

- In the 1D example, the normalization constant is easy to compute. So in practice there is no need to use MCMC samples to reconstruct the PDF, though the sample trajectory from MCMC is still useful.
- In high dimensional case, computing the normalization constant is extremely difficult, which requires solving a high-dimensional integral. Therefore, MCMC is very useful.
- In most cases, we do not know the analytic expression of the function $f(x) \propto P(x)$. Each time given an x, we can obtain an $f(x)$. This is the typically case in parameter estimation.
- In many cases, we may not even know $f(x)$, but we know $f(x')/f(x_t)$.

Using MCMC for parameter estimation

Consider the following 1D SDE,

$$du = (-au + f)dt + \sigma_u dW_u$$

where one time series is given as observation $\{u_1^{obs}, \dots u_i^{obs}, \dots u_N^{obs}\}$.

Optimization criterion: The optimal parameter correspond to the maximum of the likelihood.

Discrete form of the equation:

$$u_{i+1} = u_i + (-au_i + f)\Delta t + \sigma\sqrt{\Delta t}\epsilon, \qquad \epsilon \sim \mathcal{N}(0, 1)$$

Given the value u_i, the distribution of u_{i+1} is Gaussian,

$$p(u_{i+1}) \sim \mathcal{N}(\mu, R)$$

$$\mu = u_i + (-au_i + f)\Delta t, \qquad R = \frac{\sigma^2}{2a}(1 - e^{-2a\Delta t})$$

The likelihood is

$$L_i = p(u_{i+1}^{obs}|\theta) = \frac{1}{\sqrt{2\pi R}} \exp\left(-\frac{(u_{i+1}^{obs} - \mu)^2}{2R}\right)$$

$$L = L_1 L_2 \dots L_N$$

Log likelihood,

$$\mathcal{L} = \log(L) = \log(L_1)\log(L_2)\dots\log(L_N)$$

MCMC (Metropolis algorithm) is used to find the maximum of the log likelihood function $\mathcal{L}$, which corresponds to the optimal parameters.
In each step t,

- Sample a new set of parameters (a', f', σ') using the Metropolis algorithm, say

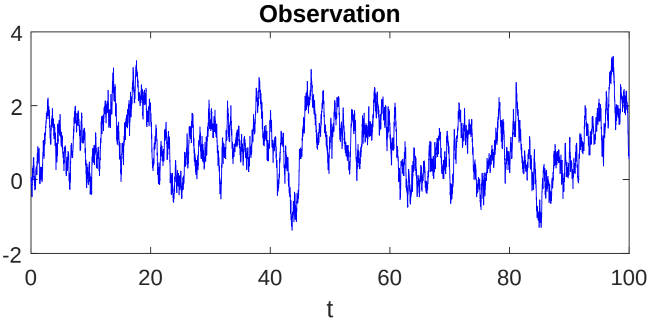
$$a' = a_t + c \cdot randn, \qquad f' = f_t + c \cdot randn, \qquad \sigma' = \sigma_t + c \cdot randn$$

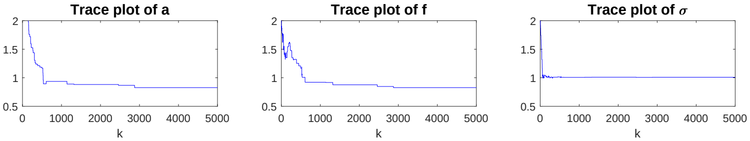
  Then calculate the log likelihood $\mathcal{L}'$ (which is the “f” function in the general MCMC discussion part).

- Calculate the acceptance ratio $\alpha = \mathcal{L}' - \mathcal{L}_t$ (think why we take the difference instead of the ratio?)

- Accept or reject the parameter candidates

$$\begin{aligned} \text{Accept:} \quad & (a_{t+1}, f_{t+1}, \sigma_{t+1}) = (a', f', \sigma') \\ \text{Reject:} \quad & (a_{t+1}, f_{t+1}, \sigma_{t+1}) = (a_t, f_t, \sigma_t) \end{aligned}$$





- Why the trace plot does not converge to the true value?
- Adaptive MCMC is often used to improve the acceptance rate.

MCMC with Data Augmentation

Consider a coupled nonlinear system

$$du = G(u, v)dt + \sigma_u dW_u,$$
$$dv = H(u, v)dt + \sigma_v dW_v.$$

Assume the observations only contain time series of u, that is, $\mathbf{u} = \{u_0^{obs}, u_1^{obs}, \ldots, u_N^{obs}\}$.
Unfortunately, different from the fully observed case, the likelihood $p(\boldsymbol{\theta}|\mathbf{u})$ has no closed analytic form. One way to handle such a system is to use the so-called data augmentation,

$$\begin{aligned} p(\boldsymbol{\theta}, v^{mis}|\mathbf{u}) &\propto p(\boldsymbol{\theta}, v^{mis}, \mathbf{u}) \\ &= p(\boldsymbol{\theta})p(v^{mis}, \mathbf{u}|\boldsymbol{\theta}) \\ &= p(\boldsymbol{\theta})p(v^{mis}|\boldsymbol{\theta})p(\mathbf{u}|v^{mis}, \boldsymbol{\theta}), \end{aligned}$$

where v^{mis} represents a full path of v.

1. Compute the auxiliary conditional distribution $p(\boldsymbol{\theta}, v^{mis}|\mathbf{u})$.
2. Marginalize over v^{mis} in $p(\boldsymbol{\theta}, v^{mis}|\mathbf{u})$ to obtain $p(\boldsymbol{\theta}|\mathbf{u})$.

Other subtle issues:

- Data augmentation is very expensive because the dimension of v^{mis} is infinite.
- The noise coefficients in the hidden processes are hard to update. Girsanov theorem needs to be applied to change the measure.
- In practice, the observations may be given at discrete points. So data augmentation needs to be applied for both v and the u between nearby observational points.

Parameter Estimation with Partial Observations Using Nonlinear Smoothing

$$\boxed{\begin{aligned} \mathrm{d}\mathbf{X}(t) &= \Big[\mathbf{A}_0(\mathbf{X},t) + \mathbf{A}_1(\mathbf{X},t)\mathbf{Y}(t)\Big]\,\mathrm{d}t + \mathbf{B}_{\mathbf{X}}(\mathbf{X},t)\,\mathrm{d}\mathbf{W}_1(t),\\ \mathrm{d}\mathbf{Y}(t) &= \Big[\mathbf{a}_0(\mathbf{X},t) + \mathbf{a}_1(\mathbf{X},t)\mathbf{Y}(t)\Big]\,\mathrm{d}t + \mathbf{b}_{\mathbf{Y}}(\mathbf{X},t)\,\mathrm{d}\mathbf{W}_2(t).\end{aligned}}$$

- For the simplicity of discussions, let us apply an Euler-Maruyama scheme to the original continuous system with a small time step Δt.
- Thus, the values of $\mathbf{X}$ and $\mathbf{Y}$ are taken at discrete points in time $\mathbf{X}(t_j)$ and $\mathbf{Y}(t_j)$, for $j = 0, 1, \ldots, J$, where $T = J\Delta t$. Define $\widehat{\mathbf{X}} = \{\mathbf{X}^0, \ldots, \mathbf{X}^j, \ldots, \mathbf{X}^J\}$ and $\widehat{\mathbf{Y}} = \{\mathbf{Y}^0, \ldots, \mathbf{Y}^j, \ldots, \mathbf{Y}^J\}$, where $\mathbf{X}^j := \mathbf{X}(t_j)$ and $\mathbf{Y}^j = \mathbf{Y}(t_j)$.
- **Parameter estimation with partial observations.** Given an ansatz of the model, the goal here is to maximize the objective function, which is the log likelihood,

$$\mathcal{L}(\boldsymbol{\theta}) = \log p(\widehat{\mathbf{X}}|\boldsymbol{\theta}) = \log \int_{\widehat{\mathbf{Y}}} p(\widehat{\mathbf{X}}, \widehat{\mathbf{Y}}|\boldsymbol{\theta})\,\mathrm{d}\widehat{\mathbf{Y}},$$

where $\boldsymbol{\theta}$ is the collection of model parameters.

Using any distribution $Q(\widehat{\mathbf{Y}})$ over the hidden variables, a lower bound on the likelihood $\mathcal{L}$ can be obtained in the following way

$$
\begin{aligned}
\log \int_{\widehat{\mathbf{Y}}} p(\widehat{\mathbf{X}}, \widehat{\mathbf{Y}}|\boldsymbol{\theta})\, \mathrm{d}\widehat{\mathbf{Y}} &= \log \int_{\widehat{\mathbf{Y}}} Q(\widehat{\mathbf{Y}}) \frac{p(\widehat{\mathbf{X}}, \widehat{\mathbf{Y}}|\boldsymbol{\theta})}{Q(\widehat{\mathbf{Y}})}\, \mathrm{d}\widehat{\mathbf{Y}} \\
&\geq \int_{\widehat{\mathbf{Y}}} Q(\widehat{\mathbf{Y}}) \log \frac{p(\widehat{\mathbf{X}}, \widehat{\mathbf{Y}}|\boldsymbol{\theta})}{Q(\widehat{\mathbf{Y}})}\, \mathrm{d}\widehat{\mathbf{Y}} \\
&= \int_{\widehat{\mathbf{Y}}} Q(\widehat{\mathbf{Y}}) \log p(\widehat{\mathbf{X}}, \widehat{\mathbf{Y}}|\boldsymbol{\theta})\, \mathrm{d}\widehat{\mathbf{Y}} - \int_{\widehat{\mathbf{Y}}} Q(\widehat{\mathbf{Y}}) \log Q(\widehat{\mathbf{Y}})\, \mathrm{d}\widehat{\mathbf{Y}} \\
&:= \mathcal{F}(Q, \boldsymbol{\theta}),
\end{aligned}
$$

where

- the negative value of $\int_{\widehat{\mathbf{Y}}} Q(\widehat{\mathbf{Y}}) \log p(\widehat{\mathbf{X}}, \widehat{\mathbf{Y}}|\boldsymbol{\theta})\, \mathrm{d}\widehat{\mathbf{Y}}$ is the so-called free energy while
- $-\int_{\widehat{\mathbf{Y}}} Q(\widehat{\mathbf{Y}}) \log Q(\widehat{\mathbf{Y}})\, \mathrm{d}\widehat{\mathbf{Y}}$ is the entropy.

Therefore, based on the fact $\mathcal{F}(Q, \boldsymbol{\theta}) \leq \mathcal{L}(\boldsymbol{\theta})$, it is clear that maximizing the log likelihood is equivalent to maximizing $\mathcal{F}$ alternatively with respect to the distribution Q and the parameters $\boldsymbol{\theta}$. This can be achieved by the expectation-maximization (EM) algorithm,

$$
\begin{aligned}
\text{E-Step:} &\qquad Q_{k+1} \leftarrow \arg\max_{Q} \mathcal{F}(Q, \boldsymbol{\theta}_k), \\
\text{M-Step:} &\qquad \boldsymbol{\theta}_{k+1} \leftarrow \arg\max_{\boldsymbol{\theta}} \mathcal{F}(Q_{k+1}, \boldsymbol{\theta}).
\end{aligned}
$$

$$\begin{aligned}\text{E-Step:} \qquad & Q_{k+1} \leftarrow \arg\max_{Q} \mathcal{F}(Q, \boldsymbol{\theta}_k), \\ \text{M-Step:} \qquad & \boldsymbol{\theta}_{k+1} \leftarrow \arg\max_{\boldsymbol{\theta}} \mathcal{F}(Q_{k+1}, \boldsymbol{\theta}).\end{aligned}$$

- The maximization in the E-Step is reached when Q is exactly the conditional distribution of $\widehat{\mathbf{Y}}$ corresponding to the **nonlinear smoother estimates**, that is,

$$Q_{k+1}(\widehat{\mathbf{Y}}) = p(\widehat{\mathbf{Y}}|\widehat{\mathbf{X}}, \boldsymbol{\theta}_k).$$

- The maximum in the M-Step is obtained by maximizing the negative of the free energy

$$\boldsymbol{\theta}_{k+1} \leftarrow \arg\max_{\boldsymbol{\theta}} \int_{\widehat{\mathbf{Y}}} p(\widehat{\mathbf{Y}}|\widehat{\mathbf{X}}, \boldsymbol{\theta}_k) \log p(\widehat{\mathbf{X}}, \widehat{\mathbf{Y}}|\boldsymbol{\theta}) \, \mathrm{d}\widehat{\mathbf{Y}}.$$

Details

Let us start with writing down the discrete approximation of the original continuous system using the Euler-Maruyama scheme,

$$
\begin{aligned}
\mathbf{X}^{j+1} &= \mathbf{X}^{j} + (\mathbf{A}_0^j + \mathbf{A}_1^j \mathbf{Y}^j)\Delta t + (\mathbf{B}_{\mathbf{X}}^j \sqrt{\Delta t})\varepsilon_{\mathbf{X}}^j,\\
\mathbf{Y}^{j+1} &= \mathbf{Y}^{j} + (\mathbf{a}_0^j + \mathbf{a}_1^j \mathbf{Y}^j)\Delta t + (\mathbf{b}_{\mathbf{Y}}^j \sqrt{\Delta t})\varepsilon_{\mathbf{Y}}^j,
\end{aligned}
\tag{1}
$$

The log likelihood function of $p(\widehat{\mathbf{X}}, \widehat{\mathbf{Y}}|\boldsymbol{\theta})$ and in turn the M-Step can be solved explicitly. Denote $\mathbf{Z}^j = (\mathbf{X}^j, \mathbf{Y}^j)^{\mathtt{T}}$. If $\mathbf{Z}$ is fully observed, then clearly the one-step log-likelihood function under the model (1) would be

$$
\mathcal{N}(\boldsymbol{\mu}^j, \mathbf{R}^j) = \widetilde{C}|\mathbf{R}^j|^{-\frac{1}{2}} \exp\left(-\frac{1}{2}(\mathbf{Z}^{j+1} - \boldsymbol{\mu}^j)^*(\mathbf{R}^j)^{-1}(\mathbf{Z}^{j+1} - \boldsymbol{\mu}^j)\right), \tag{2}
$$

where $\boldsymbol{\mu}^j = \mathbf{M}^j\boldsymbol{\theta} + \mathbf{C}^j$.
However, in the partially observed nonlinear systems considered here, the variable $\mathbf{Y}$ is unobserved. The states of $\mathbf{Y}$ are estimated from the nonlinear smoother, which means the estimation contains uncertainty. Thus, an expectation of the log-likelihood function needs to be taken, and the overall objective function becomes

$$
\min_{\boldsymbol{\theta},\mathbf{R}} \widetilde{\mathcal{L}} = \min_{\boldsymbol{\theta},\mathbf{R}} \left(\sum_j \left\langle (\mathbf{Z}^{j+1} - \mathbf{M}^j\boldsymbol{\theta} - \mathbf{C}^j)^*(\mathbf{R})^{-1}(\mathbf{Z}^{j+1} - \mathbf{M}^j\boldsymbol{\theta} - \mathbf{C}^j)\right\rangle + J\log|\mathbf{R}|\right), \tag{3}
$$

where $\langle\cdot\rangle$ denotes the expectation over the uncertain component of $\mathbf{Z}^j$, namely $\mathbf{Y}^j$.

To find the minimum of $\widetilde{\mathcal{L}}$, we aim at finding $\frac{\partial \widetilde{\mathcal{L}}}{\partial \boldsymbol{\theta}} = 0$ and $\frac{\partial \widetilde{\mathcal{L}}}{\partial \mathbf{R}} = 0$. Here

$$\widetilde{\mathcal{L}} = -J \log |\mathbf{R}^{-1}| + \mathrm{Tr}\left(\sum_j \left\langle \left(\mathbf{R}^{-1} (\mathbf{Z}^{j+1} - \mathbf{M}^j \boldsymbol{\theta} - \mathbf{C}^j)(\mathbf{Z}^{j+1} - \mathbf{M}^j \boldsymbol{\theta} - \mathbf{C}^j)^* \right) \right\rangle \right), \quad (4)$$

and therefore

$$\frac{1}{J} \frac{\partial \widetilde{\mathcal{L}}}{\partial (\mathbf{R}^{-1})} = -\mathbf{R} + \frac{1}{J} \sum_j \left\langle (\mathbf{Z}^{j+1} - \mathbf{M}^j \boldsymbol{\theta} - \mathbf{C}^j)(\mathbf{Z}^{j+1} - \mathbf{M}^j \boldsymbol{\theta} - \mathbf{C}^j)^* \right\rangle . \quad (5)$$

This gives

$$\mathbf{R} = \frac{1}{J} \sum_j \left\langle (\mathbf{Z}^{j+1} - \mathbf{M}^j \boldsymbol{\theta} - \mathbf{C}^j)(\mathbf{Z}^{j+1} - \mathbf{M}^j \boldsymbol{\theta} - \mathbf{C}^j)^* \right\rangle . \quad (6)$$

On the other hand,

$$\frac{\partial \widetilde{\mathcal{L}}}{\partial \boldsymbol{\theta}} = 2 \sum_j \left\langle (\mathbf{M}^j)^* \mathbf{R}^{-1} \mathbf{M}^j \right\rangle \boldsymbol{\theta} - 2 \sum_j \left\langle (\mathbf{M}^j)^* \mathbf{R}^{-1} (\mathbf{Z}^{j+1} - \mathbf{C}^j) \right\rangle = 0, \quad (7)$$

which implies

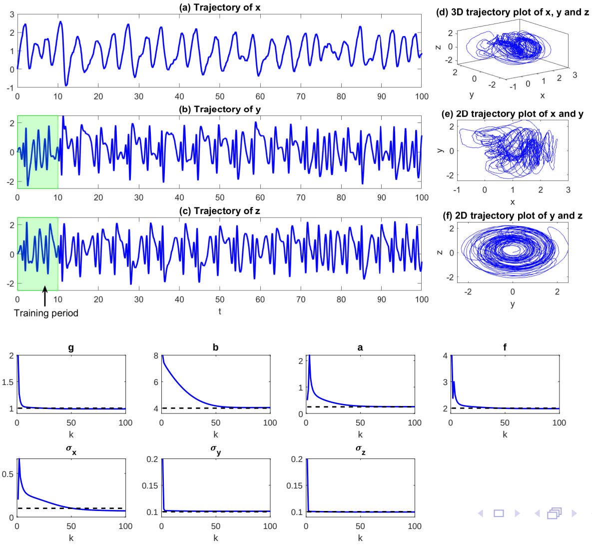
$$\boldsymbol{\theta} = \left(\sum_j \left\langle (\mathbf{M}^j)^* \mathbf{R}^{-1} \mathbf{M}^j \right\rangle \right)^{-1} \left(\sum_j \left\langle (\mathbf{M}^j)^* \mathbf{R}^{-1} (\mathbf{Z}^{j+1} - \mathbf{C}^j) \right\rangle \right). \quad (8)$$

In the derivations of (6) and (8), we have made use of the following facts,

$$\frac{\partial}{\partial \mathbf{R}} (\log |\mathbf{R}|) = \mathbf{R}^{-1} \qquad \text{and} \qquad \frac{\partial}{\partial \mathbf{R}} (\mathrm{Tr}(\mathbf{P}\mathbf{R})) = \frac{\partial}{\partial \mathbf{R}} (\mathrm{Tr}(\mathbf{R}\mathbf{P})) = \mathbf{P}^* . \quad (9)$$

Test example: The noisy Lorenz 1984 (or L84) model.

$$\begin{aligned}
dx &= \big(-(y^2 + z^2) - ax + f\big)dt + \sigma_x dW_x, \\
dy &= \big(-bxz + xy - y + g\big)dt + \sigma_y dW_y, \\
dz &= \big(bxy + xz - z\big)dt + \sigma_z dW_z.
\end{aligned} \tag{10}$$



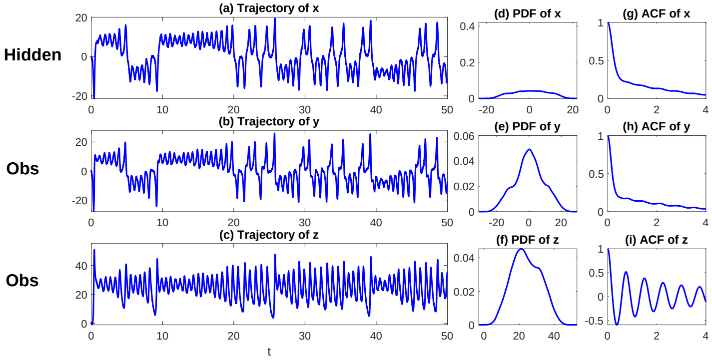
Sparse identification using the LASSO (least absolute shrinkage and selection operator) for model parsimony.

$$\min_{\boldsymbol{\theta},\mathbf{R}} \widetilde{\mathcal{L}}_{LASSO} = \min_{\boldsymbol{\theta},\mathbf{R}} \Big(\sum_{j} \Big\langle (\mathbf{Z}^{j+1} - \mathbf{M}^{j}\boldsymbol{\theta} - \mathbf{C}^{j})^{*}(\mathbf{R})^{-1}(\mathbf{Z}^{j+1} - \mathbf{M}^{j}\boldsymbol{\theta} - \mathbf{C}^{j}) \Big\rangle \\ + J \log|\mathbf{R}| \Big) + \lambda \|\boldsymbol{\theta}\|_1. \tag{11}$$

Test example: Model identification with unknown model structure.

- True signal is generated from noisy Lorenz 63 model, which is unknown.
- Given the time series of y and z, assume the following data-driven model:

$$\mathrm{d}x = (L_{11}x + L_{12}y + L_{13}z + B_{11}y^2 + B_{12}z^2 + B_{13}yz + B_{14}xy + B_{15}xz + F_1)\,\mathrm{d}t + \sigma_1\,\mathrm{d}W_1,$$
$$\mathrm{d}y = (L_{21}x + L_{22}y + L_{23}z + B_{21}y^2 + B_{22}z^2 + B_{23}yz + B_{24}xy + B_{25}xz + F_2)\,\mathrm{d}t + \sigma_2\,\mathrm{d}W_2,$$
$$\mathrm{d}z = (L_{31}x + L_{32}y + L_{33}z + B_{31}y^2 + B_{32}z^2 + B_{33}yz + B_{34}xy + B_{35}xz + F_3)\,\mathrm{d}t + \sigma_3\,\mathrm{d}W_3.$$



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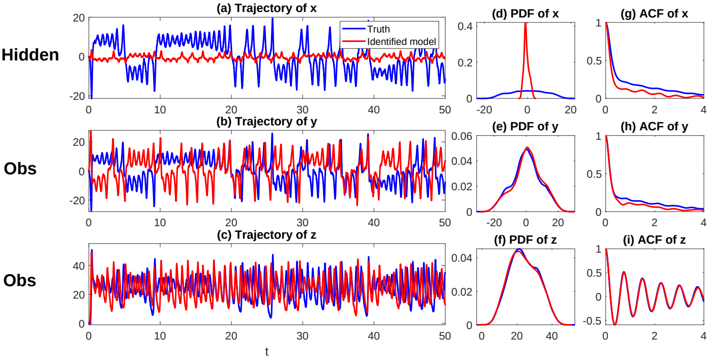
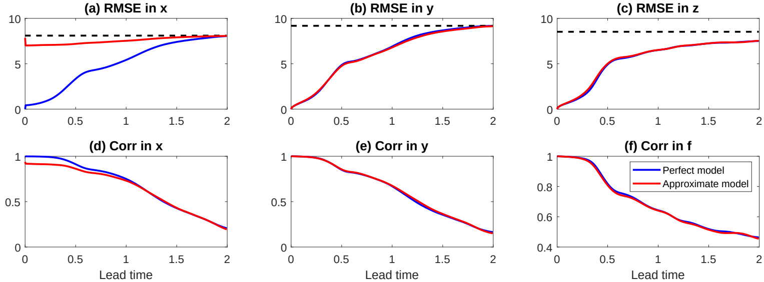
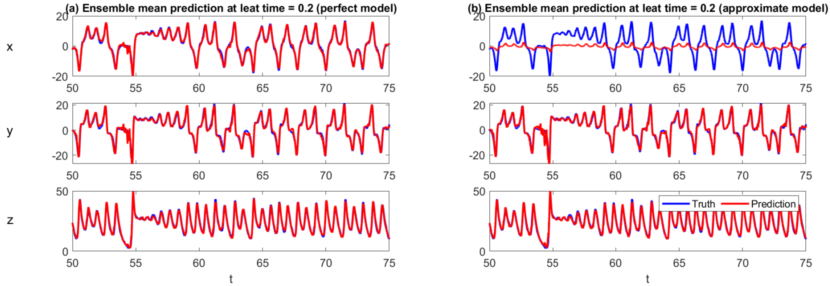
$$\mathrm{d}z = (L_{31}x + L_{32}y + L_{33}z + B_{31}y^2 + B_{32}z^2 + B_{33}yz + B_{34}xy + B_{35}xz + F_3)\,\mathrm{d}t + \sigma_3\,\mathrm{d}W_3.$$



The model learned from the partial observations (with sparse identification) can have very different behavior in the hidden variables, but the dynamical features and the prediction skill of the observed variables are identical to the perfect model.


(a) RMSE in x
(b) RMSE in y
(c) RMSE in z
(d) Corr in x
(e) Corr in y
(f) Corr in f
Perfect model
Approximate model
Lead time
Lead time
Lead time


(a) Ensemble mean prediction at leat time = 0.2 (perfect model)
(b) Ensemble mean prediction at leat time = 0.2 (approximate model)
x
y
z
Truth
Prediction
t
t

Thank you!

References and acknowledgements

1. Sanz-Alonso, Daniel, Andrew Stuart, and Armeen Taeb. Inverse problems and data assimilation. Vol. 107. Cambridge University Press, 2023.
2. Kody Law, Andrew Stuart, and Kostas Zygalakis. Data assimilation. Cham, Switzerland: Springer, 2015.
3. Andrew J Majda, and John Harlim. Filtering complex turbulent systems. Cambridge University Press, 2012.

4. We gratefully acknowledge that part of these lecture notes is adapted from Nan Chen's original materials and other online resources.